# 大学物理

电子教案

清华大学物理系基础物理教研组

# 第四章 功和能

§ 4.1 功
§ 4.2 动能定理
§ 4.3 一对力的功
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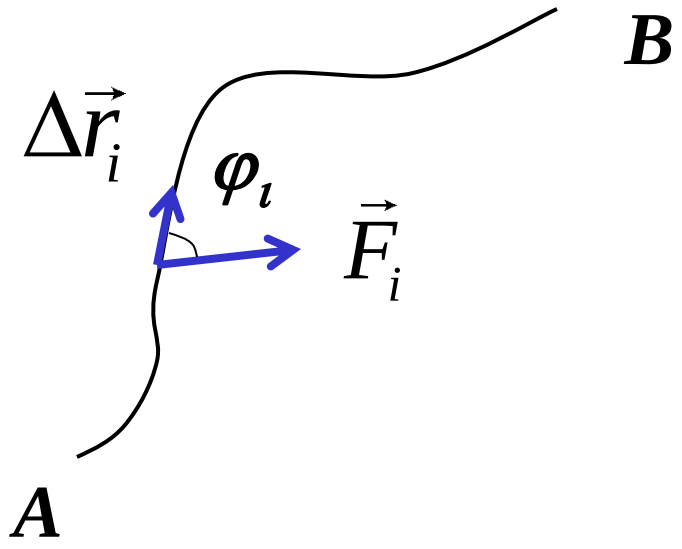
§ 4.11 伯努利方程

# § 4.1 功

## 力的空间积累




功

$$\Delta W_i = F_i \Delta r_i \cos\varphi_i$$

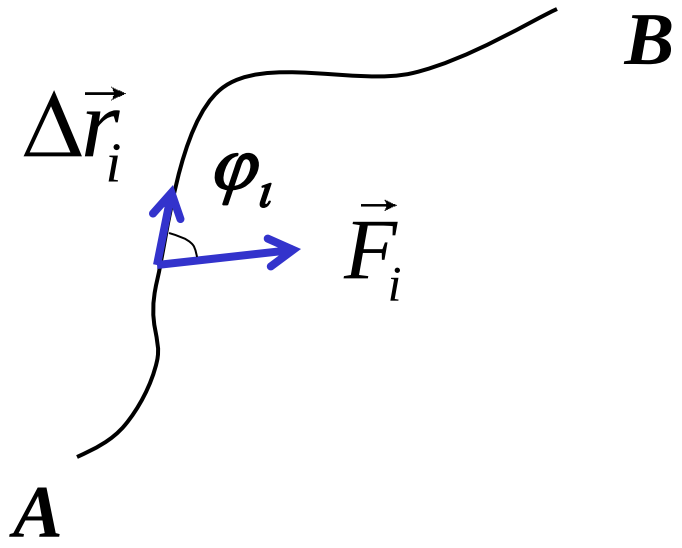
$$\Delta W_i = \vec{F}_i \cdot \Delta\vec{r}_i$$

点积（标量积）

$$\vec{A}\cdot\vec{B} = AB\cos\varphi$$

$$\vec{A}\cdot\vec{B} = A_x B_x + A_y B_y + A_z B_z$$

$$\Delta W_i = \vec{F}_i \cdot \Delta\vec{r}_i$$




**A到B做功**

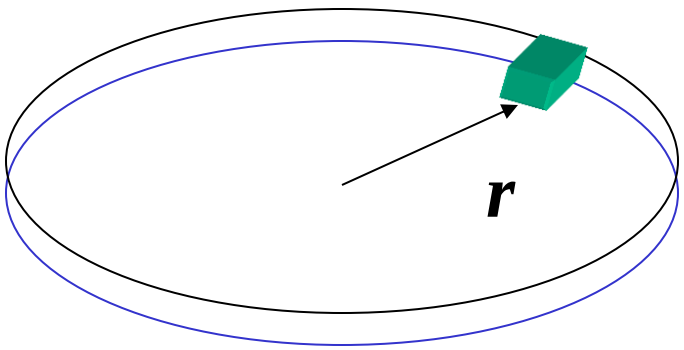
线积分

$$W_{AB} = \sum_i \Delta W_i = \sum_i \vec{F}_i \cdot \Delta\vec{r}_i = \int_A^B \vec{F} \cdot d\vec{r}$$

一般与路径有关

功率 $P_i = \vec{F}_i \cdot \dfrac{\Delta\vec{r}_i}{\Delta t} = \vec{F}_i \cdot \vec{v}_i$

例：光滑的水平桌面上有一环带，环带与小物体的摩擦系数 $\mu$，在外力作用下小物体（质量 $m$ ）以速率 $v$ 做匀速圆周运动，求转一周摩擦力做的功。




解：小物体对环带压力

$$f = m\frac{v^2}{r}$$

走一段小位移 $\Delta s$ 所做的功

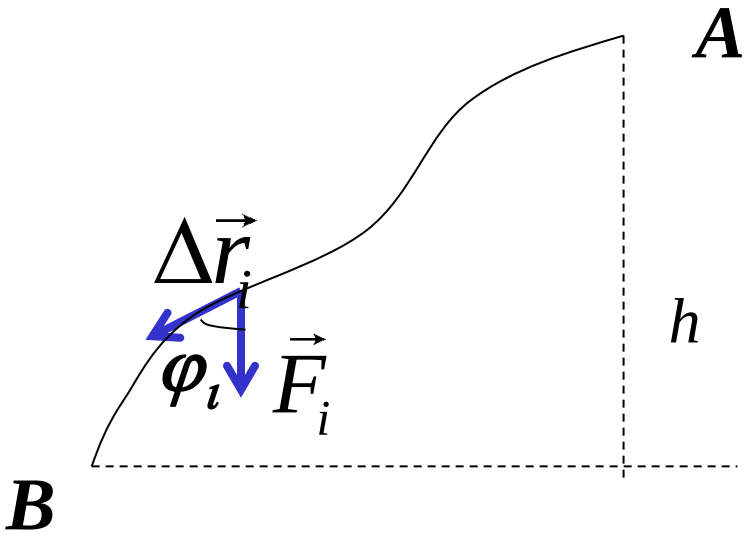
$$\Delta W = -\mu m\frac{v^2}{r}\Delta s$$

转一周

$$W = \sum_i \Delta W_i = -\mu m\frac{v^2}{r}\sum_i \Delta s = -2\pi\mu m v^2$$

# 重力做功

$$\Delta W_i = \vec{F}_i \cdot \Delta\vec{r}_i$$




**A到B做功**

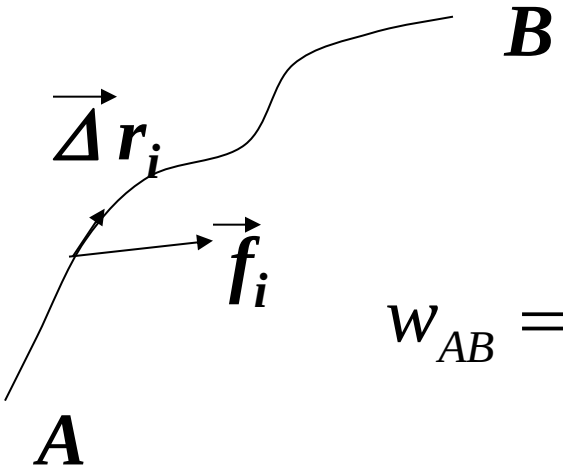
$$W_{AB} = \sum_i \Delta W_i = \sum_i \vec{F}_i \cdot \Delta\vec{r}_i = \int_A^B \vec{F} \cdot d\vec{r}$$

$$= \int_A^B -mg\hat{z} \cdot d\vec{r} = mg\int_B^A \hat{z} \cdot d\vec{r} = mgh$$

与路径无关，重力是保守力

# § 4.2 动能定理

合外力对质点所做的功等于质点动能的增量。

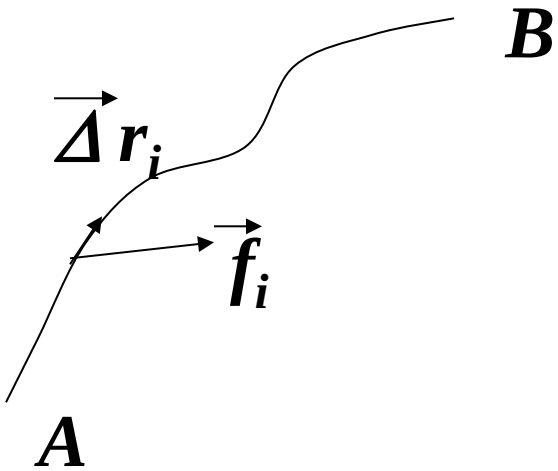



证明:

$$w_{AB} = \int_A^B \vec{f} \cdot d\vec{r} = \int_A^B m \frac{d\vec{\upsilon}}{dt} \cdot d\vec{r} = \int_A^B m d\vec{\upsilon} \cdot \vec{\upsilon}$$

$$= \int_A^B m\vec{\upsilon} \cdot d\vec{\upsilon} = \int_A^B d(\frac{1}{2} m\vec{\upsilon} \cdot \vec{\upsilon})$$

$$= \frac{1}{2} m\upsilon_B^2 - \frac{1}{2} m\upsilon_A^2 = E_{BK} - E_{AK}$$




在惯性系成立

一般地，功是路径相关的

$$w_{AB} = \int_A^B \vec{f} \cdot d\vec{r}$$

$$= E_{BK} - E_{AK}$$

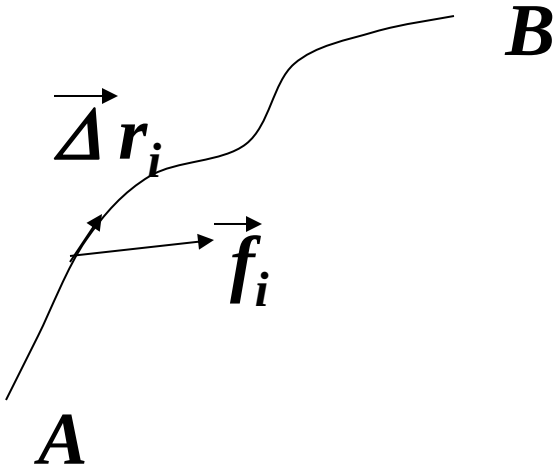
若是保守力，
则与路径无关

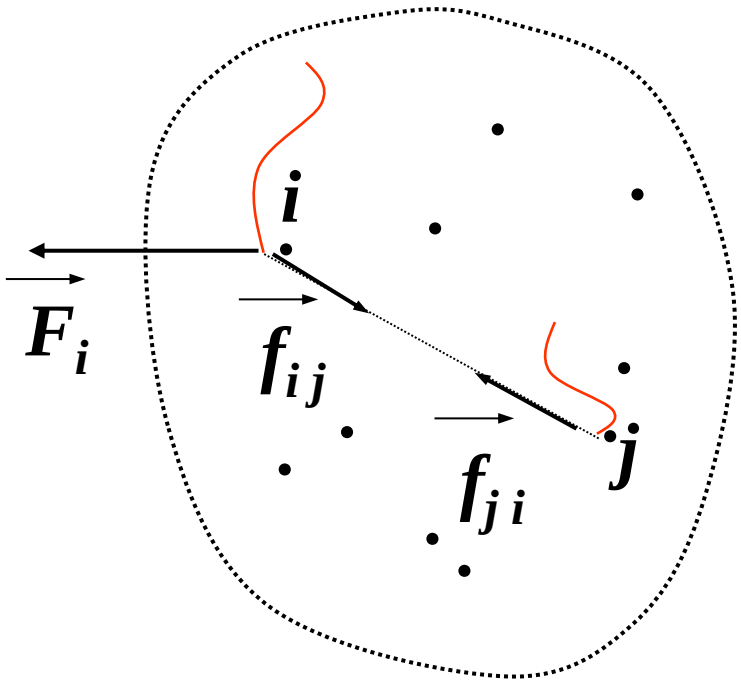
不同路径到达B，动能通常不同

$$d(\frac{1}{2}mv^2) = \vec{f} \cdot d\vec{r}$$

功率 $$\vec{f} \cdot \vec{\upsilon} = \frac{d}{dt}(\frac{1}{2}m\upsilon^2) = \frac{dE_K}{dt}$$

# 对质点系




某一时段A到B

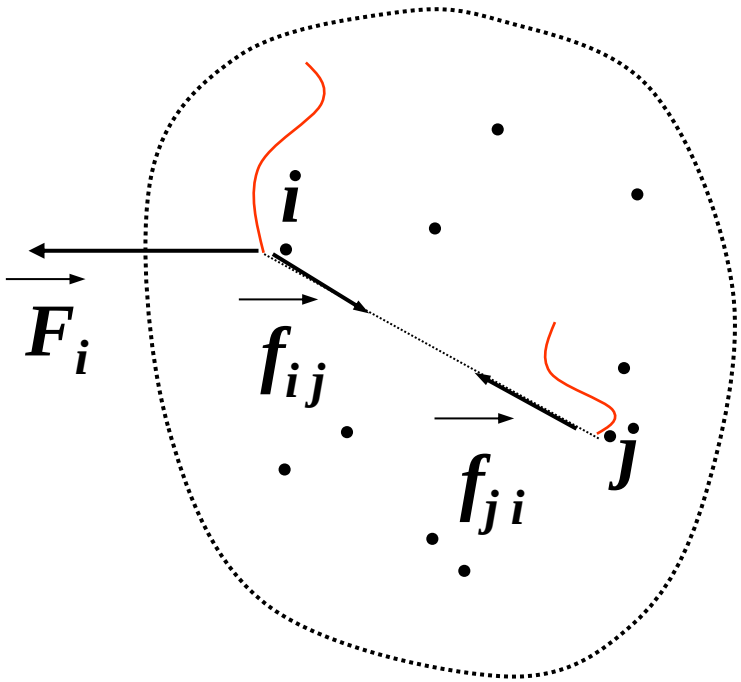
$$w_i = \int_{A_i}^{B_i} \vec{f}_i \cdot d\vec{r}_i = E_{B_i K} - E_{A_i K}$$

$$\sum_i w_i = \sum_i (E_{B_i K} - E_{A_i K})$$

$$\vec{f}_1 \cdot d\vec{r}_1 + \vec{f}_2 \cdot d\vec{r}_2$$

$$= \vec{f}_2 \cdot d(\vec{r}_2 - \vec{r}_1)$$

$$= \vec{f}_2 \cdot d\vec{r}_{21} \neq 0 \text{ (通常)}$$




内力做功和通常不为零

$$\boxed{W_{外} + W_{内} = E_{KB} - E_{KA}}$$

质点系动能定理

例：炸弹爆炸，过程内力和为零，但内力所做的功转化为弹片的动能。

※ 迄今，最不可思议的动能是，宇宙射线中有些质子的动能达到 $10^{19}$ eV，是其静止能量的$10^{10}$倍。

## 质心系

$$W'_{外,AB}+W'_{内,AB}+W'_{惯性力,AB}=E'_{KB}-E'_{KA}$$

惯性力做功

$$\sum_i -m_i\vec{a}_c\cdot d\vec{r}_i'=-\vec{a}_c\cdot d\sum_i m_i\vec{r}_i'=0$$

$$\boxed{W'_{外,AB}+W'_{内,AB}=E'_{KB}-E'_{KA}}$$

**在质心系，不论是否为惯性系，动能定理成立**

## 质心系动能

$$E'_K = \sum_i \frac{1}{2} m_i (\vec{v}_i - \vec{v}_c)^2 = \sum_i \frac{1}{2} m_i \vec{v}_i^{\,2} - \sum_i m_i \vec{v}_i \cdot \vec{v}_c + \sum_i \frac{1}{2} m_i \vec{v}_c^{\,2}$$

$$= \sum_i \frac{1}{2} m_i v_i^{\,2} - \frac{1}{2}\left(\sum_i m_i\right) v_c^{\,2} = E_K - E_{KC}$$

## 柯尼希定理

$$\boxed{E_K = E'_K + E_{KC}}$$

两体问题 $$m_1\vec{v}'_1+m_2\vec{v}'_2=0$$

相对速度 $$\vec{v}=\vec{v}_2-\vec{v}_1=\vec{v}'_2-\vec{v}'_1$$

$$E'_K=\frac{1}{2}m_1v'^2_1+\frac{1}{2}m_2v'^2_2=\frac{1}{2}\mu\vec{v}^2$$

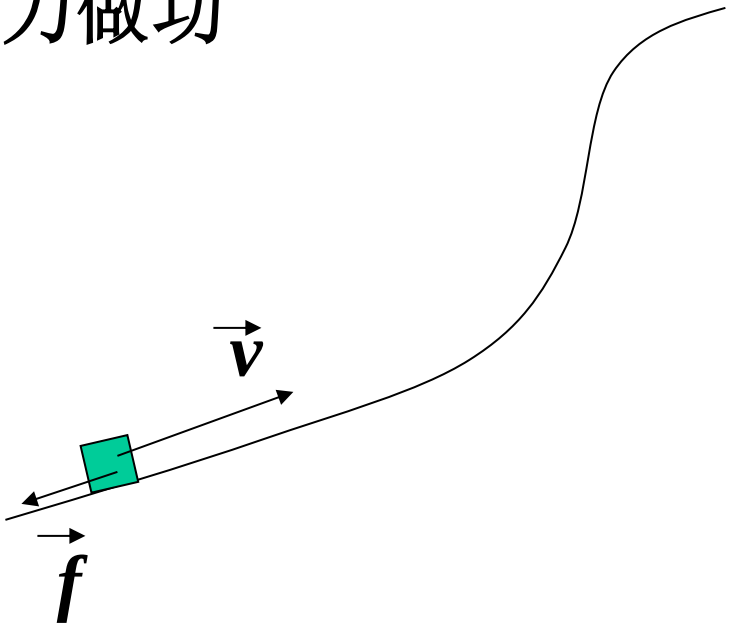
折合（约化）质量 $$\mu=\frac{m_1m_2}{m_1+m_2}$$

$$\vec{v}_1'=\vec{v}_1-\vec{V},\qquad \vec{v}_2'=\vec{v}_2-\vec{V},\qquad M=m_1+m_2$$

$$\boxed{E_K=\frac{1}{2}MV^2+\frac{1}{2}\mu v^2}$$

# § 4.3 一对力的功

例：摩擦力做功



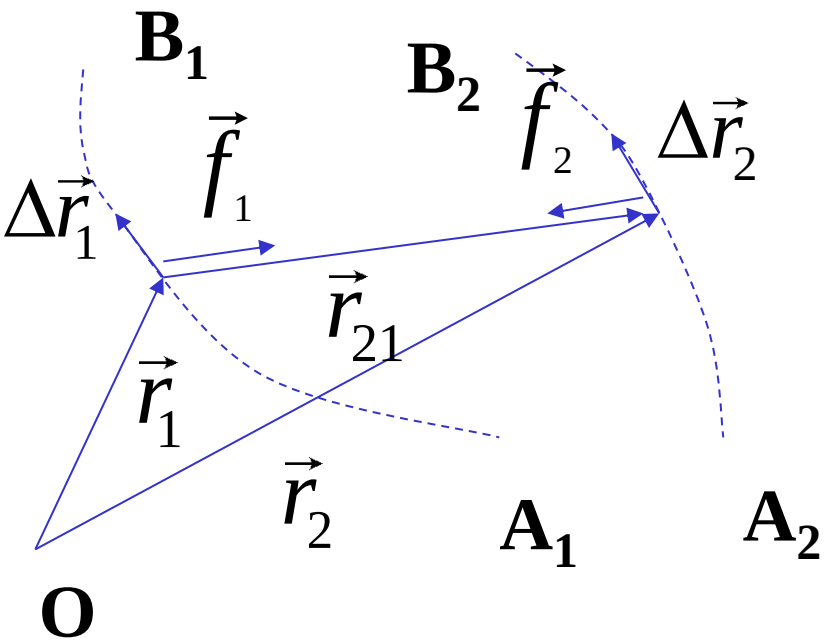

热效应

摩擦力做功 $\Delta W = -f\Delta s$

在物体参考系（也是惯性系），物体没有移动

$$\Delta W = ?$$

系统内力总是成对出现




$$\Delta W = \vec{f}_1 \cdot \Delta\vec{r}_1 + \vec{f}_2 \cdot \Delta\vec{r}_2$$

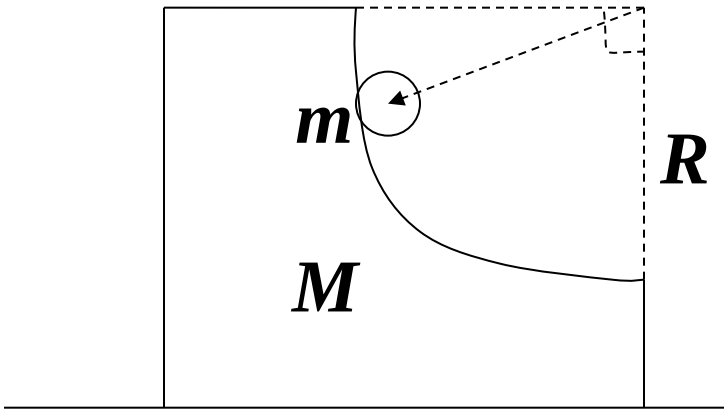
$$= \vec{f}_2 \cdot \Delta(\vec{r}_2 - \vec{r}_1) = \vec{f}_2 \cdot \Delta\vec{r}_{21}$$

一对力所做的功，等于其中一个物体所受的力沿两个物体相对移动的路径所做的功。

例题中摩擦力是一对力，
一对力所做的功与参考系的选择无关。

**例：** 有一面为1/4凹圆柱面（半径$R$）的物体（质量$M$）放置在光滑水平面，一小球（质量$m$），从静止开始沿圆面从顶端无摩擦下落（如图），小球从水平方向飞离大物体时速度 $v$，

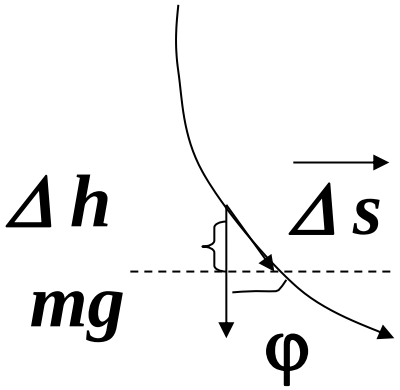
求：1）重力所做的功；2）内力所做的功。




解：重力只对小球做功

$$\Delta W_{重力} = mg\Delta s\cos\varphi = mg\Delta h$$

$$W_{重力} = mgR$$



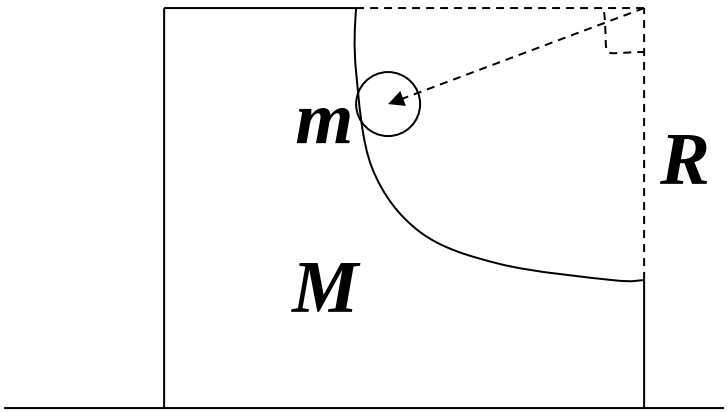

**动能定理**

$$W_{重力} + W_{内力} = \frac{1}{2}MV^2 + \frac{1}{2}m\upsilon^2$$

对$M$，内力所做的功 $\frac{1}{2}MV^2$

对$m$，内力所做的功 $\frac{1}{2}m\upsilon^2 - mgR$

内力（一对力）对两个物体分别所做功互相抵消。




$$\vec{N}_1 \cdot d\vec{r}_1 + \vec{N}_2 \cdot d\vec{r}_2$$

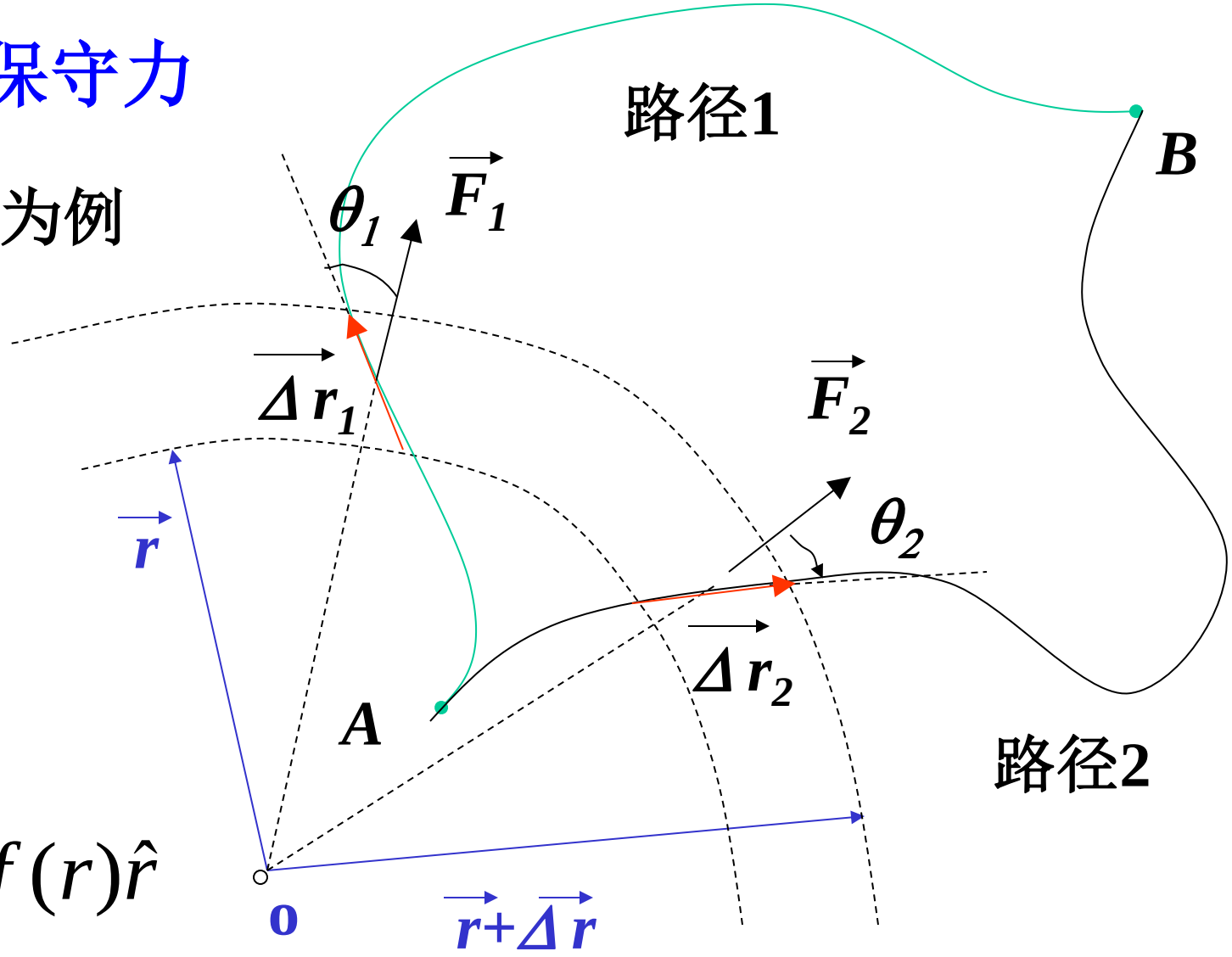
$$= \vec{N}_1 \cdot (d\vec{r}_1 - d\vec{r}_2)$$

$$= \vec{N}_1 \cdot d(\vec{r}_1 - \vec{r}_2) = 0$$

$$\frac{1}{2}MV^2 + \frac{1}{2}m\upsilon^2 = mgR = \frac{1}{2}\mu(\upsilon - V)^2$$

# § 4.4 保守力

有心力为例




$$\vec{F}(\vec{r}) = f(r)\hat{r}$$

$$\vec{F}_1 \cdot \Delta\vec{r}_1 = \vec{F}_2 \cdot \Delta\vec{r}_2 = f(r)\Delta r$$

有心力做功与路径无关

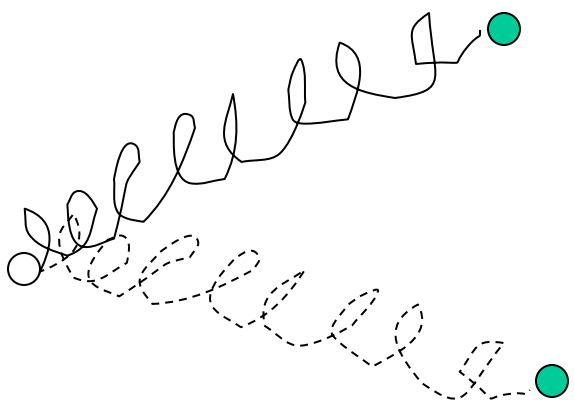
$$W_{A\to B}(1)=W_{A\to B}(2)$$

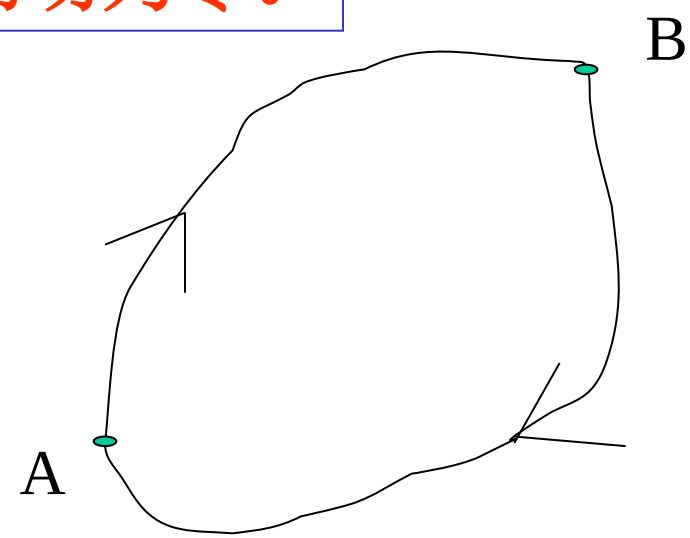
所做的功与路径无关，这种力称为保守力。

$$W_{A\to B}(2)=-W_{B\to A}(2) \qquad W_{A\to B}(1)+W_{B\to A}(2)=0$$

保守力沿任意闭合路径所做的功为零。

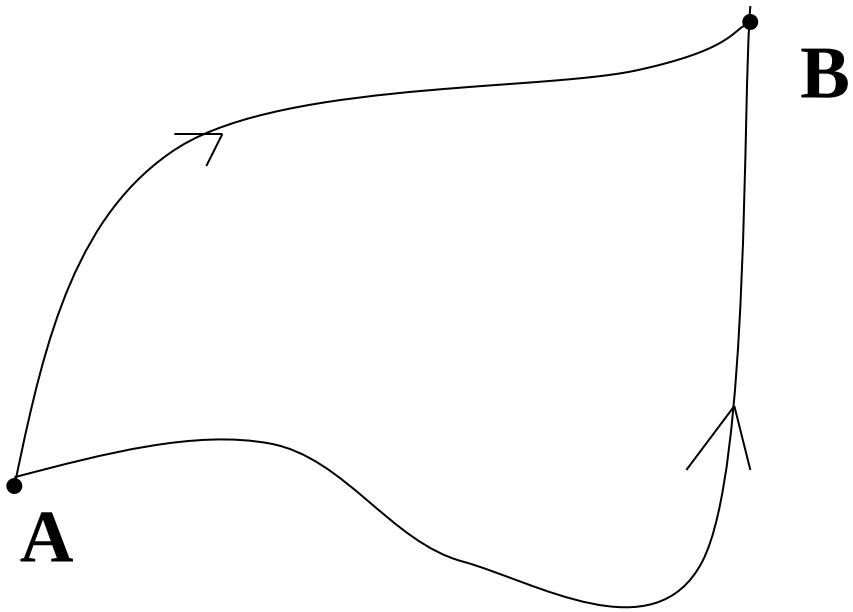
万有引力、静电力、弹性力






与保守力相对的称为耗散力，如摩擦力。

# § 4.5 势能




在保守力场（在任意点受保守力的作用），质点从A-->B，所做的功与路径无关，而只与这两点的位置有关。

可引入一个只与位置有关的函数， A 点的函数值减去(?) B点的函数值，定义为从A --> B保守力所做的功，该函数就是势能函数。

$$E_P(A) - E_P(B) = W_{A\to B}$$

定义了势能差

沿力的方向势能减小

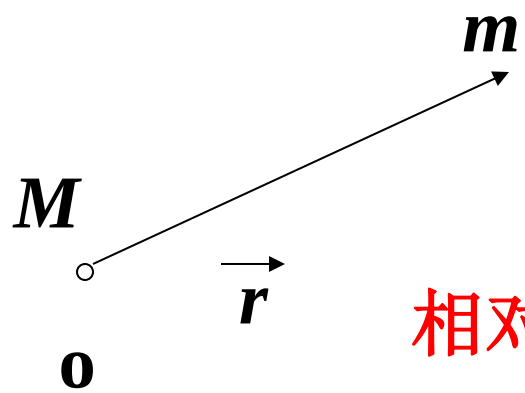
选参考点（势能零点）

设 $E_P(B) = 0$

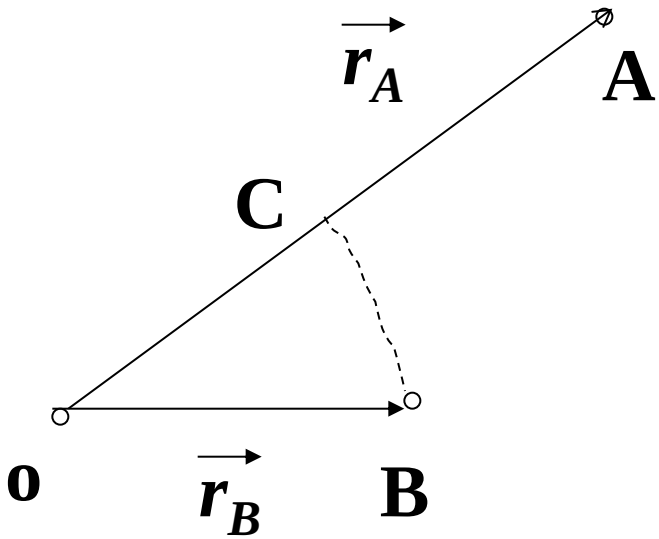
$$E_P(A) = W_{A\to B}$$

# § 4.6 万有引力势能

$$\vec{f} = -G\frac{mM}{r^2}\hat{r}$$




相对坐标




$$W_{B\to A} = W_{B\to C} + W_{C\to A} = W_{C\to A}$$

$$W_{C\to A} = \int_C^A \vec{f}\cdot d\vec{r} = -GmM\int_{r_C}^{r_A}\frac{dr}{r^2}$$

$$= -GmM\left(\frac{1}{r_C} - \frac{1}{r_A}\right)$$

相对移动　　一对力的功

$$W_{B\to A}=W_{C\to A}=GmM\left(\frac{1}{r_A}-\frac{1}{r_C}\right)=\frac{GmM}{r_A}-\frac{GmM}{r_B}$$

$$\boxed{E_P(B)-E_P(A)=W_{B\to A}}$$

两体共有的势能

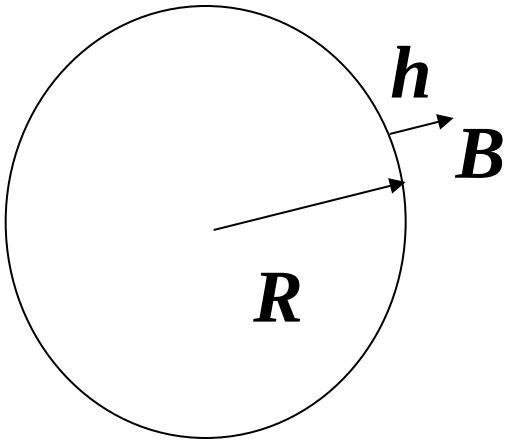
选无限远点势能为零 $r_A\to\infty$

$$\boxed{E_P=-G\frac{mM}{r}}$$

地面

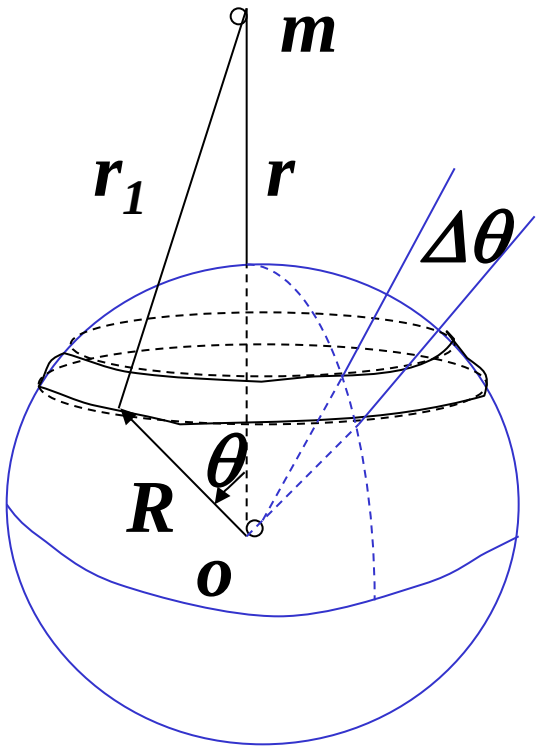
$$E_{PB}-E_{P地}=-G\frac{mM}{R+h}-\left(-G\frac{mM}{R}\right)$$

$$=\frac{GmMh}{R(R+h)}\approx m\frac{GM}{R^2}h=mgh$$




地面为势能零点

## *球壳与质点的引力




面密度 $\sigma$ $\quad M = 4\pi R^2\sigma$

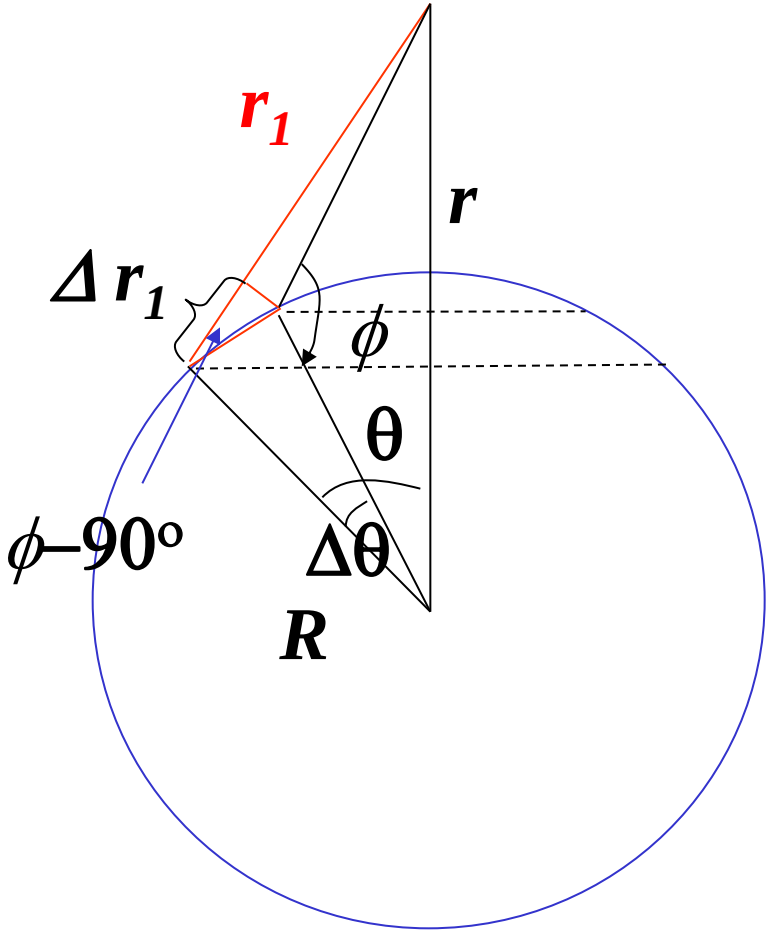
球壳上一环，宽 $R\Delta\theta$

$$M_{环} = 2\pi R\sin\theta \cdot R\Delta\theta \cdot \sigma$$

$$E_{P环} = -G\frac{m\cdot\sigma 2\pi R^2\sin\theta\Delta\theta}{r_1}$$

$$\frac{\sin\phi}{r} = \frac{\sin\theta}{r_1}$$

$$E_{P环} = -G\frac{m\cdot\sigma 2\pi R^2\Delta\theta\sin\phi}{r}$$




$$\Delta r_1 = R\Delta\theta\sin\phi$$

$$E_{P环} = -G\frac{m\cdot\sigma 2\pi R\Delta r_1}{r}$$

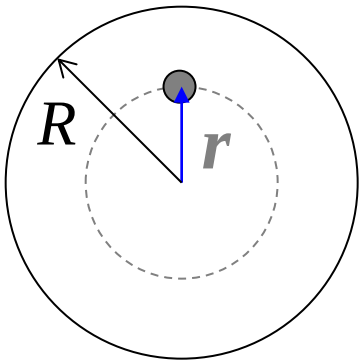
$$r_1 \in (r-R \rightarrow r+R)$$

$$\sum \Delta r_1 = (r+R)-(r-R) = 2R$$

$$E_{P壳} = -G\frac{m\cdot\sigma 2\pi R\sum\Delta r_1}{r} = -G\frac{mM}{r}$$

与壳质量全部集中在球心一样

如，质点在壳内 $r_1 \in (R-r \to r+R)$




$$\sum \Delta r_1 = (r+R)-(R-r) = 2r$$

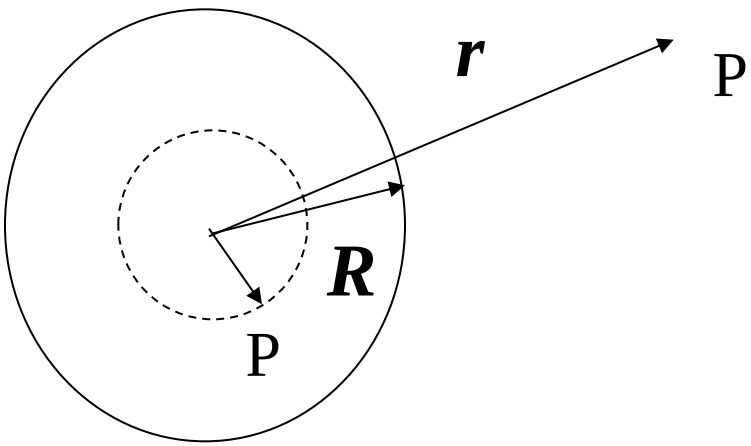
$$E_{P壳} = -G\frac{m\cdot\sigma 2\pi R\sum \Delta r_1}{r} = -G\frac{mM}{R}$$

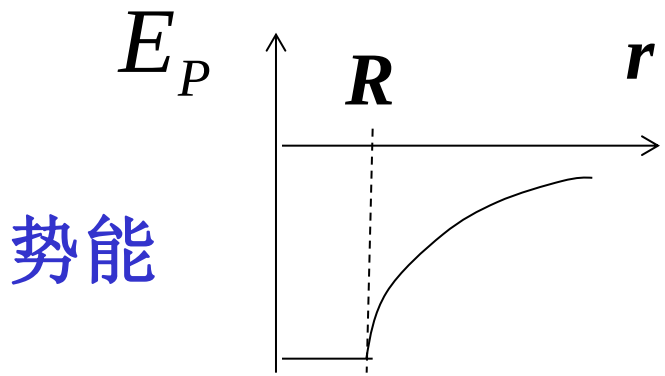
质点在壳内任意移动，
保守力不做功 -----> 质点在壳内不受力

如果是球体：质点在球内，可认为质点所在球面内的质量全部集中在球心，质点所在球面外的质量对质点无作用。

平方反比力的特征，对静电力结论正确。

结论




$r > R$

$$E_P = -G\frac{mM}{r}$$

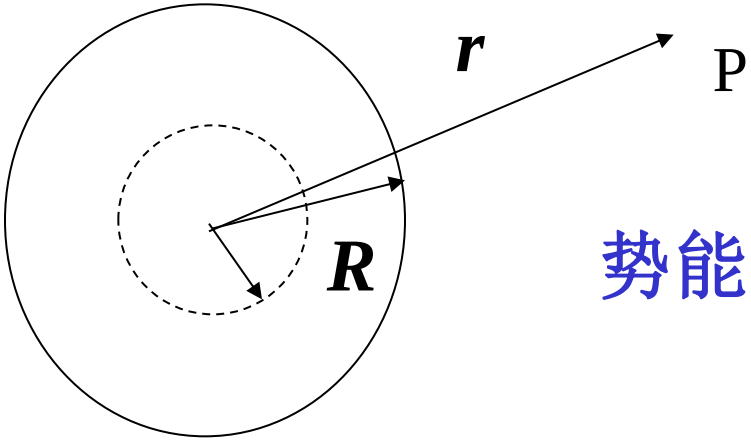
质点在球面外时, 与壳质量全部集中在球心一样

$r < R$

$$E_{P壳} = -G\frac{mM}{R}$$

质点在球面内时，壳质量对质点无作用

球体




势能

$$r > R$$

$$E_P = -G\frac{mM}{r}$$

$$r < R$$

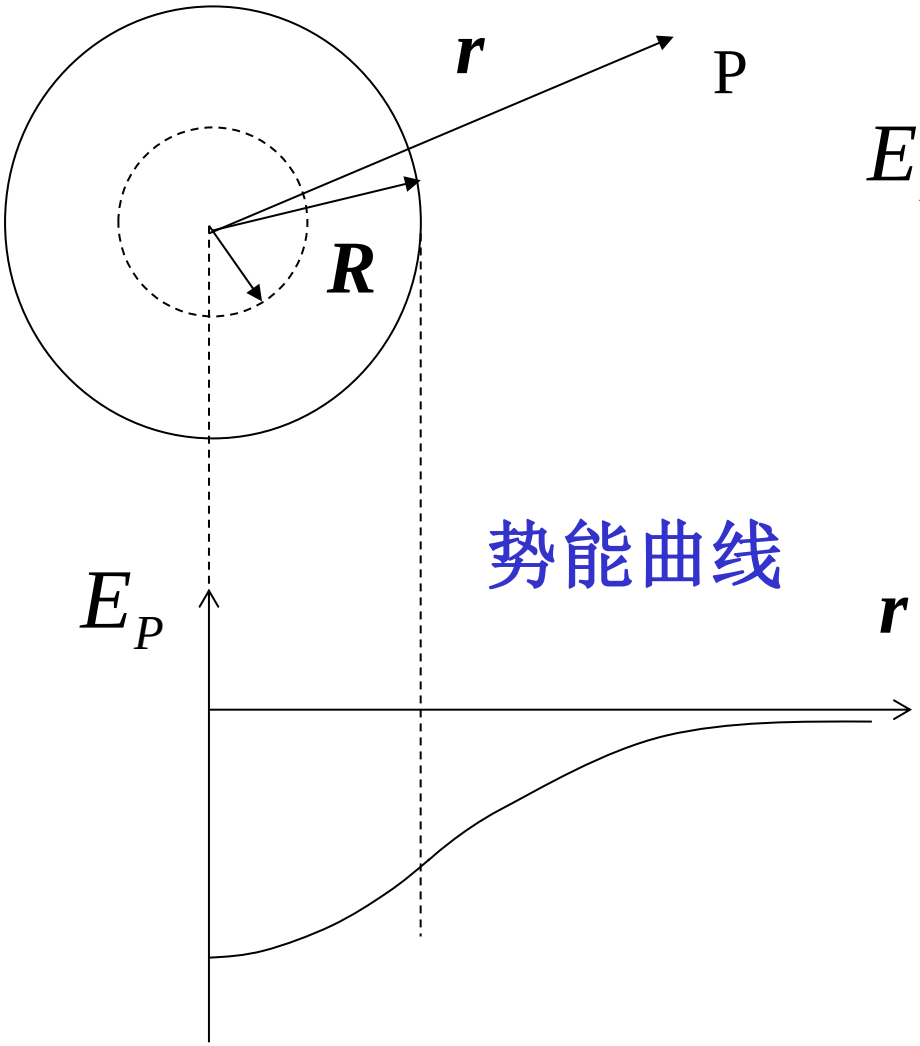
$$E_P = -G\frac{m}{r}\frac{4\pi}{3}r^3\rho - Gm\int_r^R \frac{\rho 4\pi r^2 dr}{r}$$

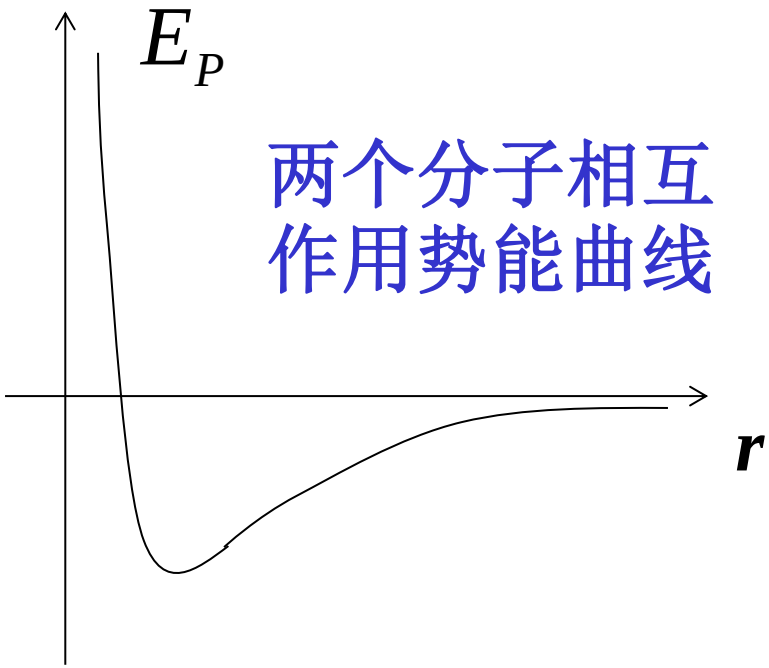
$$= \frac{1}{2}G\frac{mM}{R}\left(\frac{r^2}{R^2} - 3\right)$$

$$E_P = -G\frac{mM}{r} \qquad r > R$$

$$E_P = \frac{1}{2}G\frac{mM}{R}\left(\frac{r^2}{R^2} - 3\right)$$

$$r < R$$



# 开普勒问题

$$\ddot{\vec{r}} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + \frac{1}{r}\frac{d}{dt}(r^2\dot{\theta})\hat{\theta}$$

$$-G\frac{mM}{r^2} = m(\ddot{r} - r\dot{\theta}^2) \qquad r^2\dot{\theta} = h \qquad \dot{\theta} = \frac{h}{r^2}$$

用两体问题处理更严格

$$\boxed{\begin{gathered} -G\frac{mM}{r^2} = \mu(\ddot{r} - r\dot{\theta}^2) \\ M \to M + m \end{gathered}}$$

$$\ddot{r} - \frac{h^2}{r^3} + G\frac{M}{r^2} = 0$$

解微分方程　　轨迹是圆锥曲线

$$E = \frac{1}{2} m\dot{r}^2 + V_{eff}$$

$$V_{eff} = -G\frac{Mm}{r} + \frac{mh^2}{2r^2}$$

$$\dot{r} = \pm\sqrt{\frac{2}{m}(E - V_{eff})}$$

$$d\theta = \pm\frac{h}{r^2}\frac{dr}{\sqrt{\frac{2}{m}(E - V_{eff})}}$$

$$d\theta = \pm\frac{d(\frac{1}{r})}{\sqrt{\frac{e^2}{p^2} - (\frac{1}{r} - \frac{1}{p})^2}}$$

$$p = \frac{h^2}{GM}$$

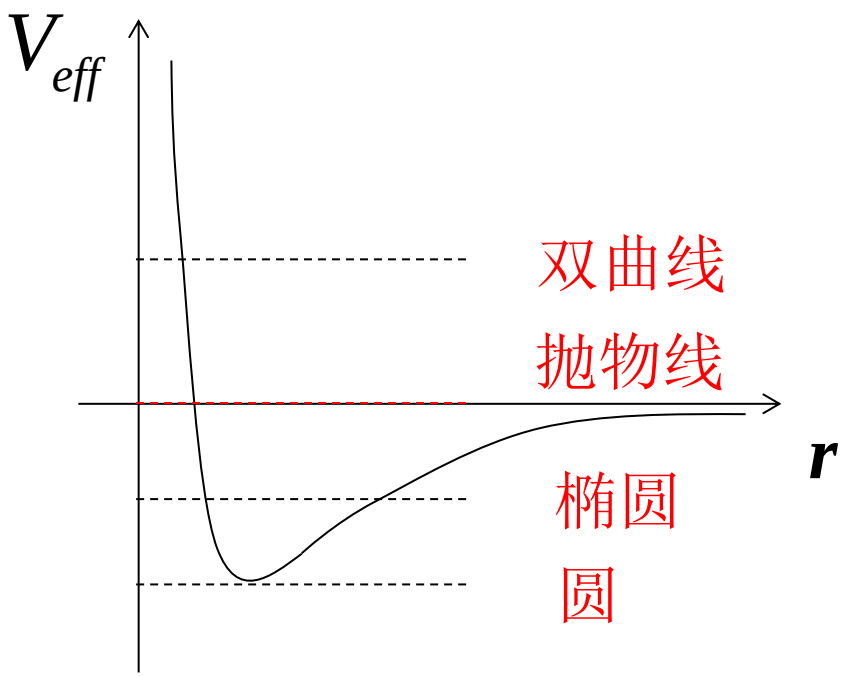
$$\theta = \pm\arccos(\frac{1}{r} - \frac{1}{p})\frac{p}{e} + c$$

$$e = \sqrt{1 + \frac{2h^2E}{G^2M^2m}}$$

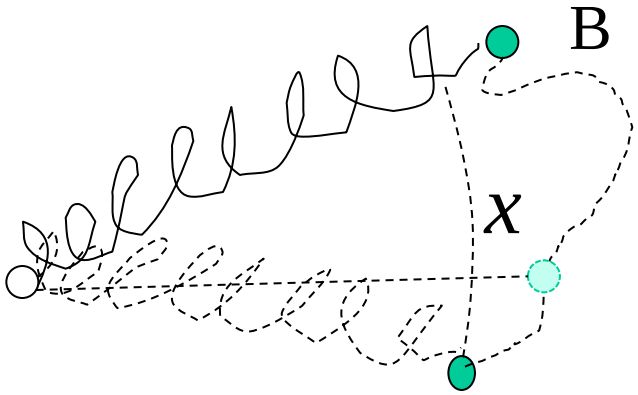
$$r = \frac{p}{1 + e\cos\theta}$$

$$e = \sqrt{1 + \frac{2h^2 E}{G^2 M^2 m}}$$

| | |
|---|---|
| $e = 0$ | 圆 |
| $0 < e < 1$ | 椭圆 |
| $e = 1$ | 抛物线 |
| $e > 1$ | 双曲线 |



# § 4.7 弹性势能




弹性力 $-kx$

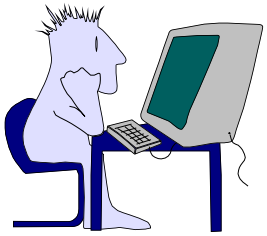
$$W_{A\to B}=\sum_i -kx_i\Delta x_i=-k\sum_i x_i\Delta x_i$$

$$W_{A\to B}=-k\int_A^B xdx=-\frac{1}{2}k(x_B{}^2-x_A{}^2)$$

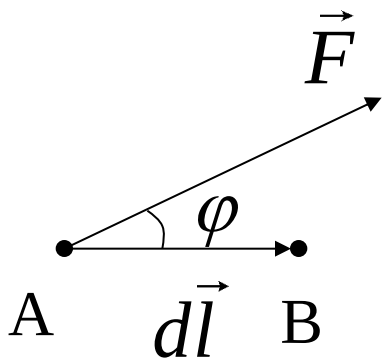
$$E_P(A)-E_P(B)=W_{A\to B}$$

**自然长度 $x_B=0$，弹性势能为零**

$$\boxed{E_P=\frac{1}{2}kx^2}$$



# § 4.8 由势能求保守力



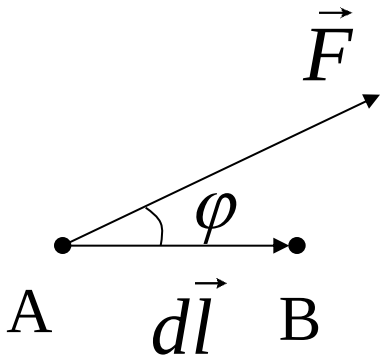

$$E_P(A) - E_P(B) = W_{A\to B}$$

$$-dE_P = W_{A\to B} = \vec{F}\cdot d\vec{l} = F_l dl$$

$$\boxed{F_l = -\frac{dE_P}{dl}}$$

$$E_P = -G\frac{mM}{r} \quad \to \quad F_r = -\frac{dE_P}{dr} = -G\frac{mM}{r^2}$$

$$E_P = \frac{1}{2}kx^2 \quad \to \quad F_x = -\frac{dE_P}{dx} = -kx$$




$$F_l = F\cos\varphi$$

$$F_l = -\frac{dE_P}{dl}$$ ***l*** 方向变化率

$$\boxed{F_l = -\frac{\partial E_P}{\partial l}}$$ ***l*** 方向导数

$E_P$ 沿哪个方向的变化率最大？ $\hat{F}$ 方向

梯度：变化最陡的方向导数

$$\boxed{\vec{F} = -grad(E_P)}$$

***F*** = 负的沿 $\hat{F}$ 方向 $E_P$ 的变化率

例 $$E_P = -G\frac{mM}{r} \quad \rightarrow \quad \vec{F} = -grad(E_P)$$

变化最陡的方向导数

$$\vec{F} = -\frac{\partial E_P}{\partial r}\hat{r} = -G\frac{mM}{r^2}\hat{r}$$

*x* 分量 $$F_x = -\frac{\partial E_P}{\partial x}$$

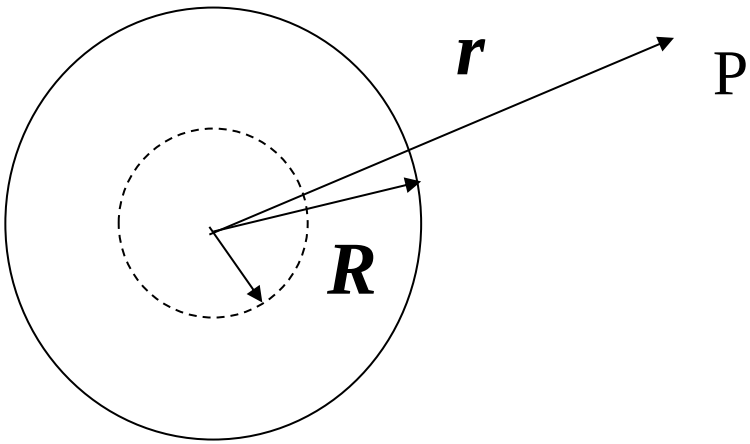
同理对 *y*, *z* 分量

$$\vec{F} = F_x\vec{i} + F_y\vec{j} + F_z\vec{k}$$

$$= -\vec{i}\frac{\partial E_P}{\partial x} - \vec{j}\frac{\partial E_P}{\partial y} - \vec{k}\frac{\partial E_P}{\partial z}$$

$$\boxed{\vec{F} = -\nabla E_P}$$

$$\boxed{\nabla = \vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}}$$




$$E_P = -G\frac{mM}{r} \qquad r > R$$

$$E_P = \frac{1}{2}G\frac{mM}{R}\left(\frac{r^2}{R^2} - 3\right)$$

$$r < R$$

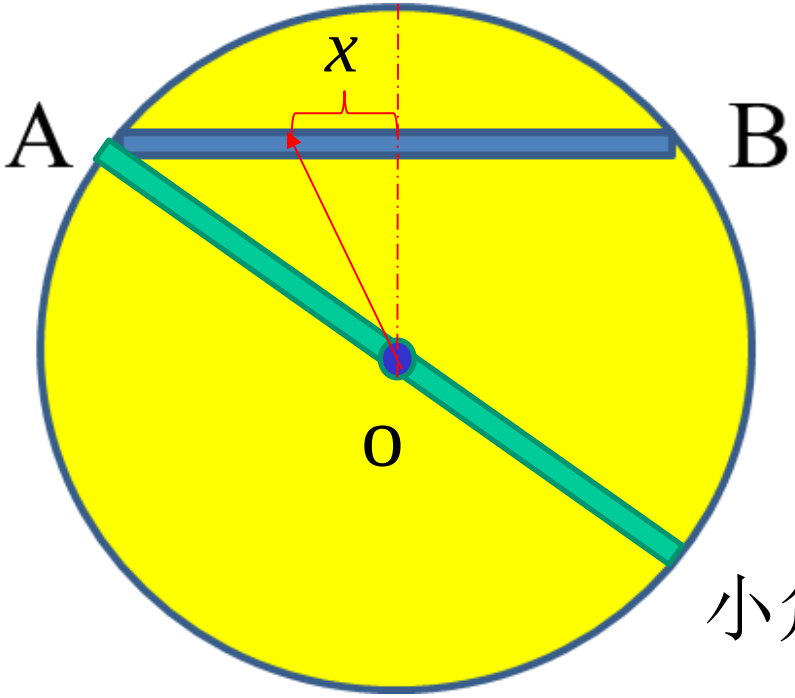
力

$$\vec{F} = -\nabla E_P \rightarrow F_r = -\frac{dE_P}{dr}$$

$$\vec{F} = -\nabla E_P = -\frac{\partial E_P}{\partial r}\hat{r} = -G\frac{mM}{R^3}\vec{r} \qquad r < R$$

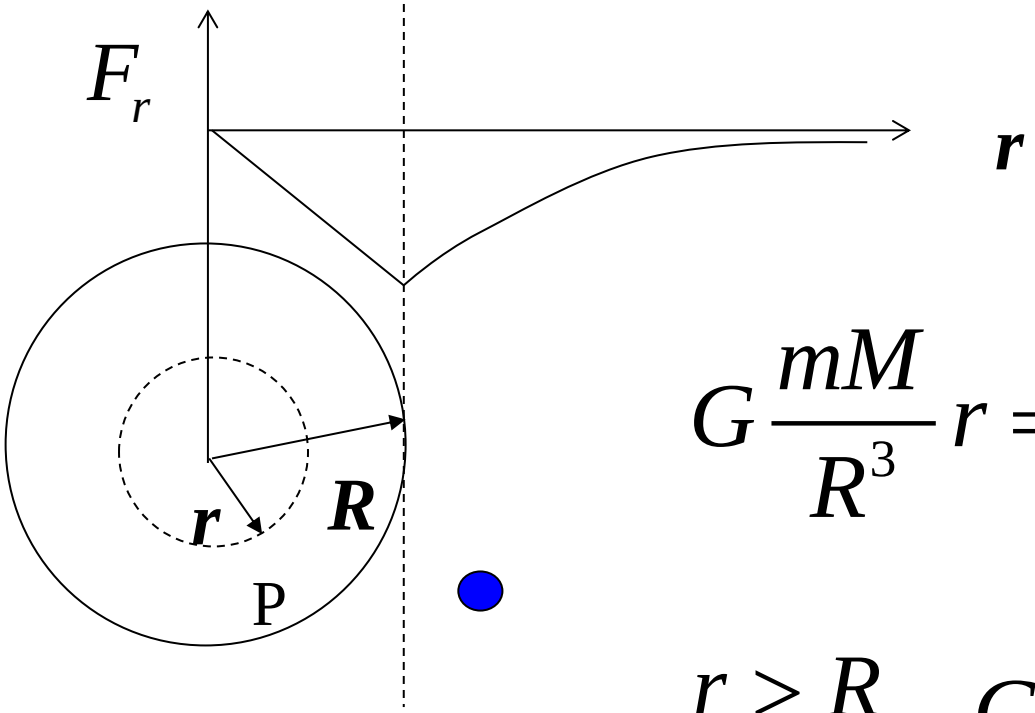
$$= -G\frac{mM}{r^2}\hat{r} \qquad r > R$$

$$\vec{F} = -G\frac{mM}{R^3}\vec{r} \qquad r < R$$




小角度单摆或弹簧

$$F_x = -G\frac{mM}{R^3}r\frac{x}{r} = -G\frac{mM}{R^3}x$$



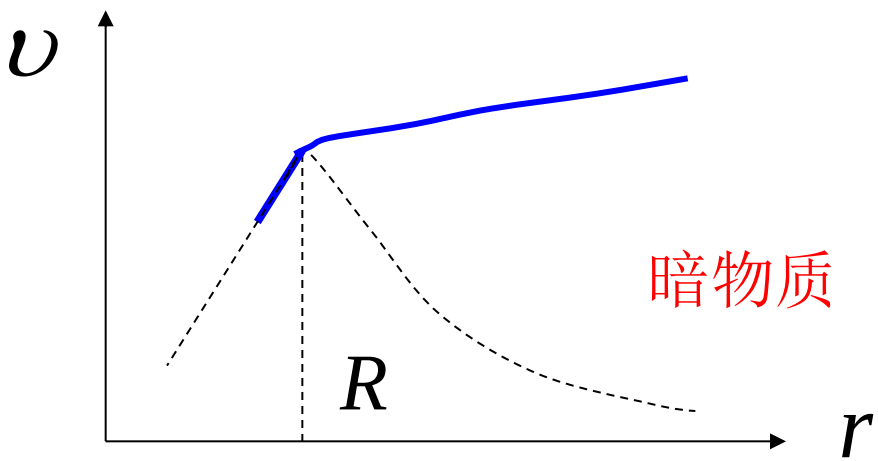

# 恒星运动速度

$$G\frac{mM}{R^3}r = m\frac{\upsilon^2}{r}$$

$$\boxed{\upsilon \propto r}$$

$$r > R \qquad G\frac{mM}{r^2} = \frac{m\upsilon^2}{r}$$

$$\boxed{\upsilon \propto \frac{1}{\sqrt{r}}}$$



# § 4.9 机械能守恒定律

$$W_{外}+W_{内}=E_{KB}-E_{KA}$$ 内力分为两部分

$$W_{外}+W_{内非}+W_{内保}=E_{KB}-E_{KA}$$

$$W_{内保A\to B}=E_P(A)-E_P(B)$$

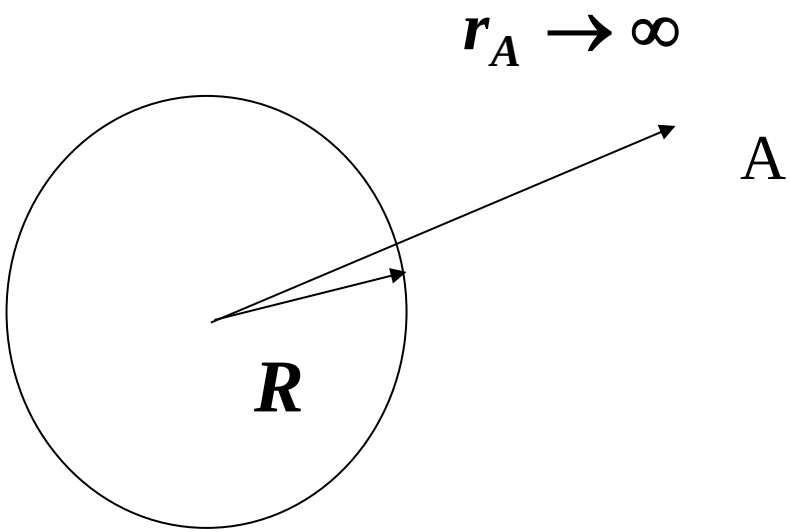
$$\underbrace{W_{外}+W_{内非}=(E_{KB}+E_{PB})-(E_{KA}+E_{PA})}_{\text{功能原理}}$$

机械能 $E_K+E_P$

质点系只有保守内力做功，机械能守恒。

更普遍地，孤立系统能量守恒。

# 逃逸速度




$$-G\frac{mM}{R}+\frac{1}{2}m\upsilon_e^2=0$$

$$\boxed{\upsilon_e=\sqrt{\frac{2GM}{R}}=\sqrt{2gR}}$$

经典黑洞 $\boxed{\upsilon_e=c}$

$$\boxed{R=\frac{2GM}{c^2}}$$

例：在平面两相同的球做非对心完全弹性碰撞，其中一球开始时处于静止状态，另一球速度 $\vec{v}$。

求证：碰撞后两球速度总互相垂直。

解：设碰撞后两球速度 $\vec{v}_1, \vec{v}_2$

由动量守恒 $\vec{v} = \vec{v}_1 + \vec{v}_2$

两边平方 $v^2 = v_1^2 + 2\vec{v}_1 \cdot \vec{v}_2 + v_2^2$

由机械能守恒（势能无变化） $v^2 = v_1^2 + v_2^2$

$\vec{v}_1 \cdot \vec{v}_2 = 0$ 两球速度总互相垂直

## 守恒律与对称性

动量守恒：　空间平移对称性

角动量守恒：空间各向同性对称性

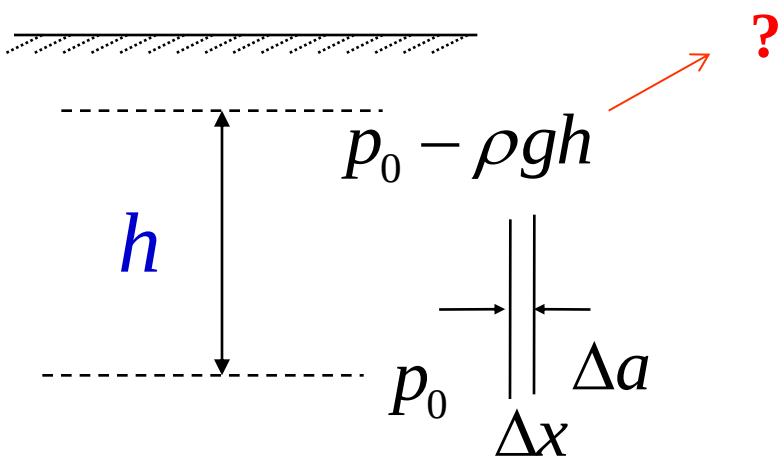
能量守恒：　时间平移对称性

诺特定理：对称性对应物理量守恒

# § 4.10 流体的稳定流动

流体：特殊的质点系，可以流动

静流体没有切向力, 只有压强 (各向同性)




单位质量势能 $U$

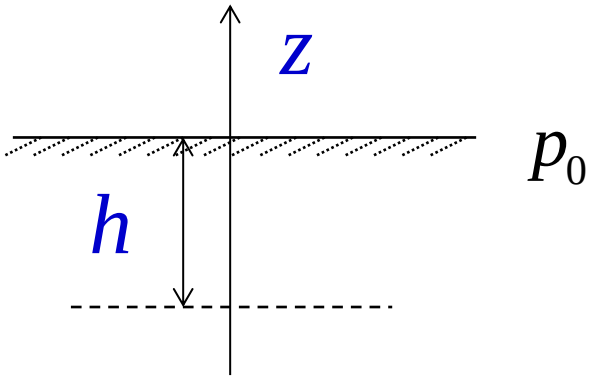
$$-\frac{\partial p}{\partial x} - \rho \frac{\partial U}{\partial x} = 0$$

$$-[p(x+\Delta x) - p(x)]\Delta a - \Delta a \Delta x \rho \frac{\partial U}{\partial x} = 0$$ 同理对 ***y, z*** 分量

$$\boxed{-\nabla p - \rho \nabla U = 0}$$

密度不变 $\boxed{p + \rho U = const.}$

重力作用下 $p + \rho g z = const.$




$$p_0 = p - \rho g h$$

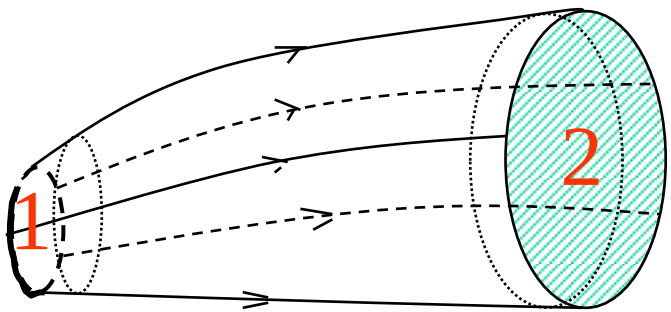
地球中心压强?

运动流体非常复杂, 如湍流

流体有粘滞性

一般液体密度变化不大，假设是常量
且忽略粘滞, 就是理想流体 (模型）

流场速度不随时间改变--- 稳定流动




$$\rho \upsilon_1 A_1 \Delta t = \rho \upsilon_2 A_2 \Delta t$$

$$\upsilon_1 A_1 = \upsilon_2 A_2$$

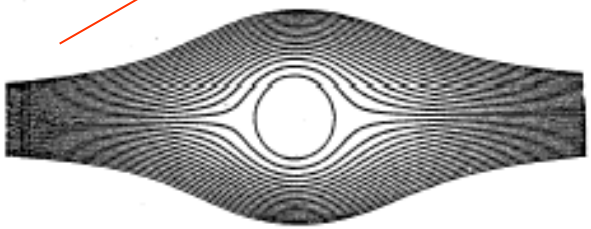
# 真实流体

稳定流动

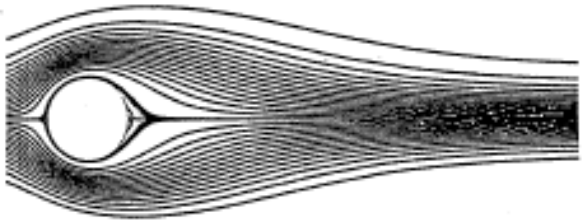
雷诺数

$$\mathrm{Re} = \frac{\rho V D}{\eta}$$

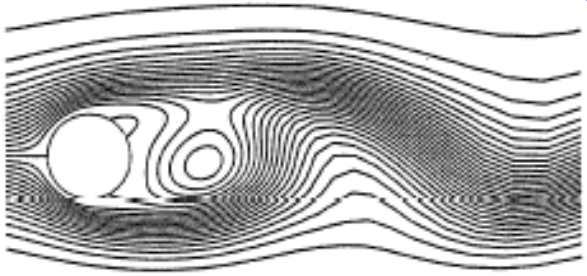
乒乓球 ~30m/s  Re ~ 5000



(a) $Re = 0.1$

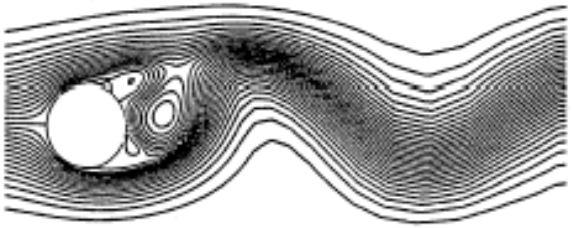


(b) $Re = 7.0$

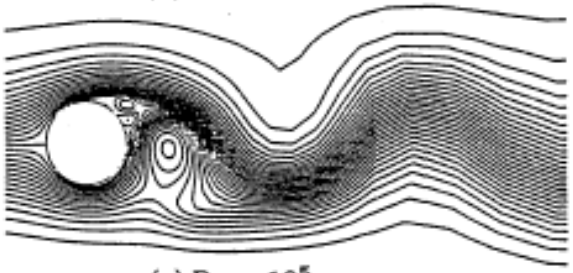


(c) $Re = 10^2$

湍流

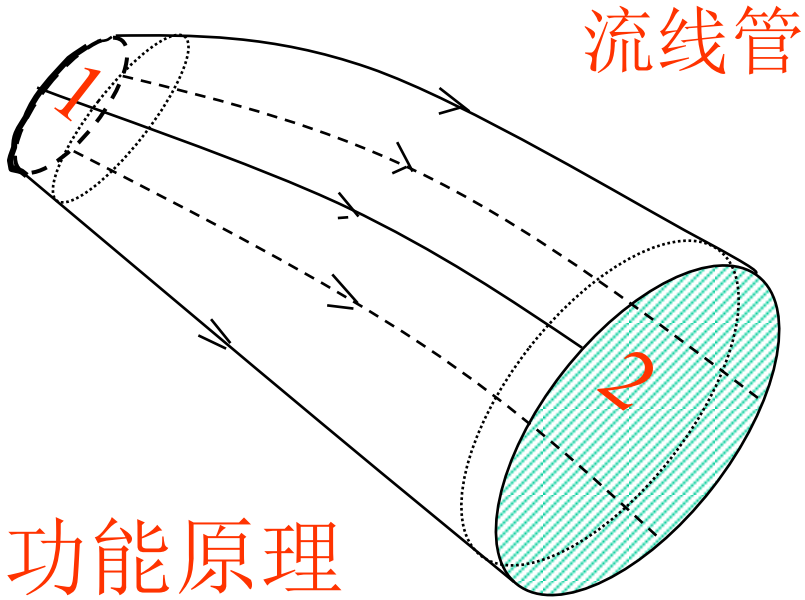


(d) $Re = 10^4$



(e) $Re = 10^5$

# § 4.11 伯努利方程




$$E = \frac{1}{2}\upsilon^2 + U$$

$$\Delta M = \rho_1 \upsilon_1 A_1 \Delta t = \rho_2 \upsilon_2 A_2 \Delta t$$

功能原理

$$p_1 A_1 \upsilon_1 \Delta t - p_2 A_2 \upsilon_2 \Delta t = \Delta M (E_2 - E_1)$$

$$E_2 - E_1 = \frac{p_1 A_1 \upsilon_1 \Delta t}{\Delta M} - \frac{p_2 A_2 \upsilon_2 \Delta t}{\Delta M} = \frac{p_1}{\rho_1} - \frac{p_2}{\rho_2}$$

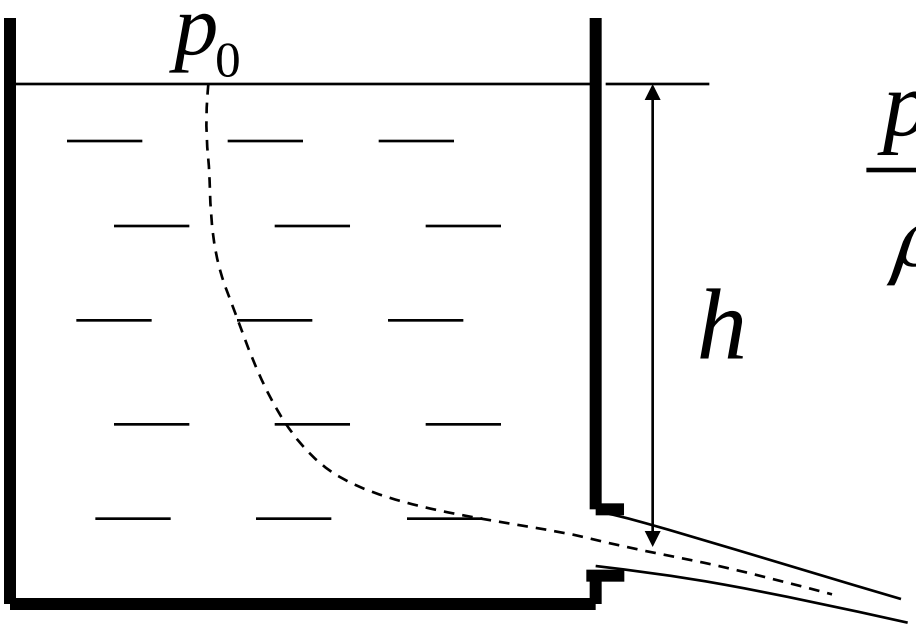
$$E_2 + \frac{p_2}{\rho_2} = E_1 + \frac{p_1}{\rho_1}$$

沿流线 $$\boxed{\frac{p}{\rho} + \frac{1}{2}\upsilon^2 + U = const.}$$

**伯努利方程**

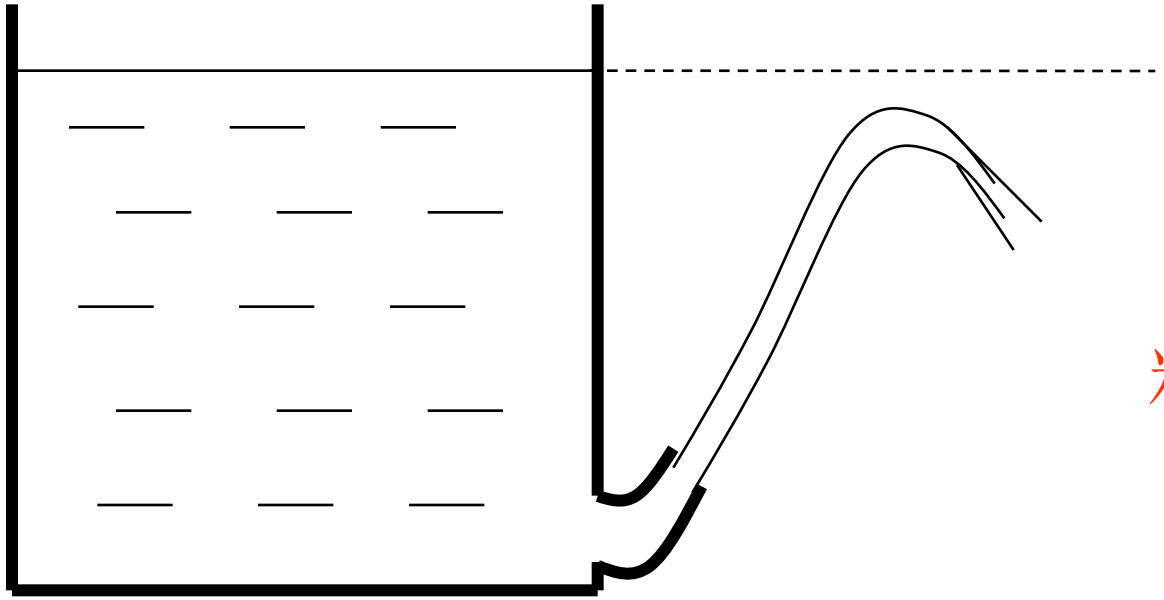
若无旋 $$\boxed{\frac{p}{\rho} + \frac{1}{2}\upsilon^2 + U = const.}$$

不仅仅沿流线, 处处是常量

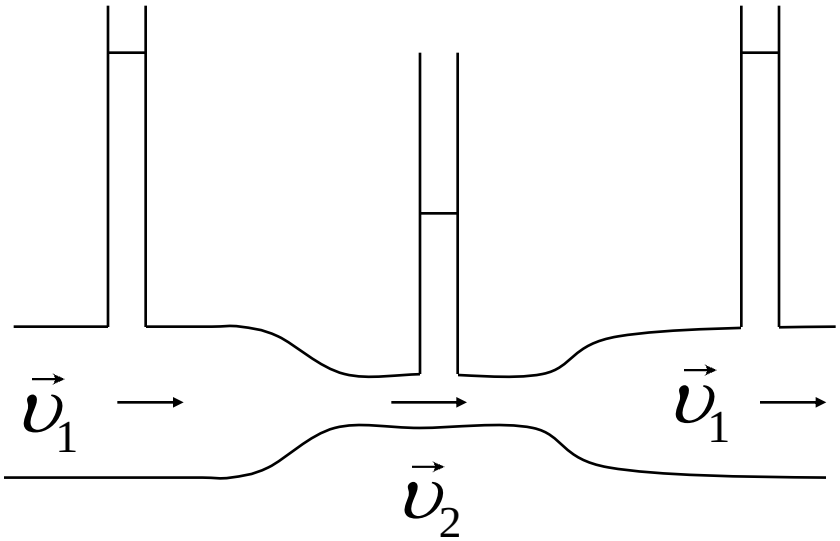



$$\frac{p_0}{\rho} = \frac{p_0}{\rho} + \frac{1}{2}\upsilon^2 - gh$$

$$\upsilon = \sqrt{2gh}$$

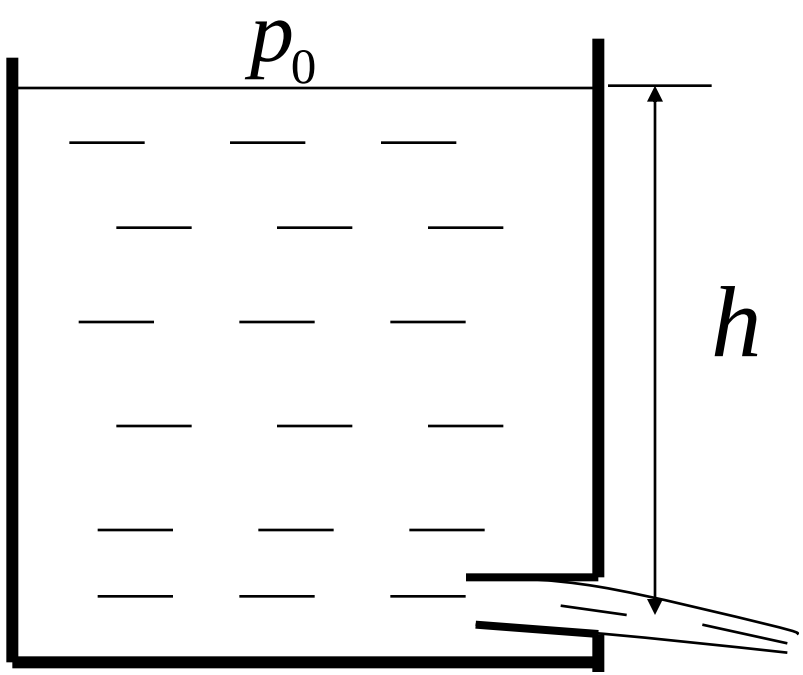


粘滞等




流速大处压强小

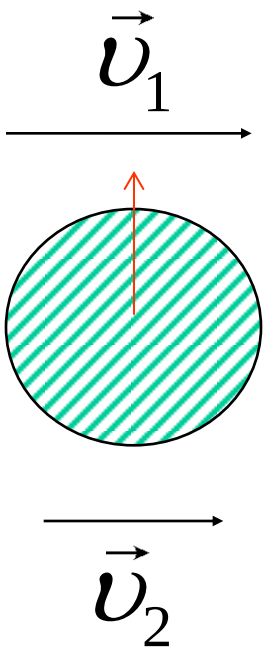
$$\frac{p}{\rho}+\frac{1}{2}\upsilon^2+U=const.$$




水流收缩

$$\rho g h A_{hole}=\rho \upsilon A_{out}\upsilon$$

$$A_{out}=\frac{1}{2}A_{hole}$$




# Bernoulli's theorem

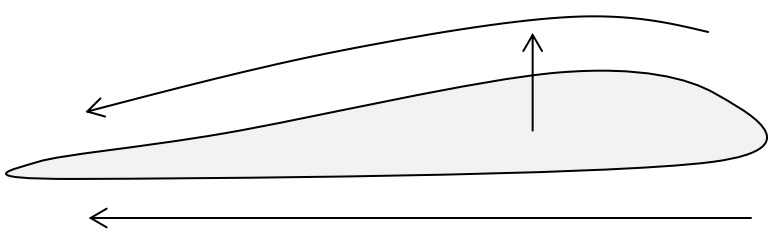
$$\frac{p}{\rho} + \frac{1}{2}\upsilon^2 + U = const.$$

$$\upsilon_1 > \upsilon_2 \qquad p_1 < p_2$$

## Lift force

飞机机翼的升力

香蕉球?



第四章完

本章编者

安宇